

DAVID E. V. ROSE

September 28, 2026

RESEARCH INTERESTS

- Low-dimensional topology and TQFT
- Representation theory
- Homological algebra and category theory

EDUCATION

- 2012 Duke University**, Durham, NC
Ph.D. in Mathematics
Advisor: Lenny Ng
Thesis: *Categorification of Quantum $\mathfrak{sl}_3$ Projectors and the $\mathfrak{sl}_3$ Reshetikhin-Turaev Invariant of Framed Tangles*
- 2007 Christ's College, University of Cambridge**, Cambridge, UK
Certificate of Advanced Studies (Part III) with Merit
Essay: *Dirac Operators*
- 2006 The College of William and Mary**, Williamsburg, VA
B.S. in Mathematics and Physics, *Summa Cum Laude* with highest (research) honors
Mathematics Advisor: Ilya Spitkovsky
Mathematics Thesis: *Results concerning the Aluthge transform*
Physics Advisor: Christopher Carone
Physics Thesis: *Minimal length uncertainty and the quantum mechanics of non-commutative space-time*

PROFESSIONAL EXPERIENCE

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|--------------------------------|--|
| July 2022 – | University of North Carolina , Chapel Hill, NC
Associate Professor
(On research and study leave: Fall 2026)
(On parental leave: Spring 2023) |
| July 2016 – June 2022 | University of North Carolina , Chapel Hill, NC
Assistant Professor
(On research and study leave: Fall 2018) |
| August 2012 – July 2016 | University of Southern California , Los Angeles, CA
Busemann Assistant Professor (NTT postdoc) |

HONORS

- Senior Fellow, **Institute for Pure and Applied Mathematics (IPAM)**, Los Angeles, CA. Long Program on *Quantum topology, character varieties and low-dimensional geometry*, Fall 2026
- **G. de B. Robinson Award** of the Canadian Mathematical Society, 2024
- **NSF CAREER Award**, 2022
- Invited researcher and group co-leader, **American Institute of Mathematics**, San Jose, CA. *Link homology research community*, Summer and Fall 2021.
- Invited researcher, **Kavli Institute for Theoretical Physics**, Santa Barbara, CA. Program on *Quantum Knot Invariants and Supersymmetric Gauge Theories*, Fall 2018.
- Invited researcher, **Isaac Newton Institute**, Cambridge, UK. Program on *Homology theories in low dimensional topology*, Spring 2017.
- **Bass Fellowship** for designing and teaching the undergraduate course *Algebraic methods in knot theory* at Duke University, Dates: 9/1/2011 – 5/31/2012
- **L.P. and Barbara Smith award for beginning teachers** in the Duke University Mathematics Department, February, 2009
- **Cambridge overseas trust scholarship**, September, 2006
- **William and Mary Prize in Mathematics**, awarded to the top graduating Mathematics major, May 2006
- **Phi Beta Kappa**, College of William and Mary, November, 2005
- **E. G. Clark memorial scholarship**, awarded to the top junior in the William and Mary Department of Physics, May 2005
- **Monroe Scholar**, College of William and Mary, Fall 2002 – Spring 2006

PUBLICATIONS AND PREPRINTS

Authors on papers are listed alphabetically (as is convention in pure mathematical writing) and papers are listed reverse-chronologically within section, in order of publication/completion. Refereed publications are those that have been published (or accepted for publication) as of September 28, 2026. A dagger[†] denotes a collaborator who was a student during the collaboration.

Refereed publications:

1. E. Bodish[†], B. Elias, D.E.V. Rose, and L. Tatham[†]. **Type C webs**. 2026. Accepted for publication at *Communications of the American Mathematical Society*. 47 pages.
2. M. Hogancamp, D.E.V. Rose, and P. Wedrich. **A skein relation for singular Soergel bimodules**. *Selecta Mathematica, New Series*. 32, 35 (2026). <https://doi.org/10.1007/s00029-025-01121-x>. 47 pages.
3. E. Bodish[†], B. Elias, D.E.V. Rose, and L. Tatham. **Decreasing subsequences and Viennot for oscillating tableaux**. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*. 95, 161–179 (2025). <https://doi.org/10.1007/s12188-026-00295-1>
4. M. Hogancamp, D.E.V. Rose, and P. Wedrich. **A Kirby color for Khovanov homology**. *Journal of the European Mathematical Society (JEMS)*. (2025), published online first DOI 10.4171/JEMS/1589. 60 pages.

5. M. Abram[†], L. Lamberto-Egan[†], A. Lauda, and D.E.V. Rose. **Categorification of the internal braid group action for quantum groups I: 2-functoriality**. *Pacific Journal of Mathematics*, Vol. 328 (2024), No. 1, 1–75.
6. M. Hogancamp, D.E.V. Rose, and P. Wedrich. **Link splitting deformation of colored Khovanov–Rozansky homology**. *Proceedings of the London Mathematical Society* (3) 129 (2024), no. 3, Paper No. e12620, 142 pages.
7. D.E.V. Rose and L. Tatham[†]. **On webs in quantum type C**. *Canadian Journal of Mathematics*, 74(3), June 2022, 793–832. (For this paper, we were awarded the **G. de B. Robinson Award** by the CMS.)
8. D.E.V. Rose and D. Tubbenhauer. **HOMFLYPT homology for links in handlebodies via type A Soergel bimodules**, *Quantum Topology*, 12 (2021), no. 2, 373–410.
9. H. Queffelec and D.E.V. Rose. **Sutured annular Khovanov–Rozansky homology**. *Transactions of the AMS*, 370 (2018), 1285–1319.
10. D.E.V. Rose and P. Wedrich. **Deformations of colored $\mathfrak{sl}_n$ link homologies via foams**. *Geometry and Topology*, 20 (2016), no. 6, 3431–3517.
11. D.E.V. Rose and D. Tubbenhauer. **Symmetric webs, Jones–Wenzl recursions, and q -Howe duality**. *International Mathematics Research Notices*, 2016 (17): 5249–5290.
12. H. Queffelec and D.E.V. Rose. **The $\mathfrak{sl}_n$ foam 2-category: a combinatorial formulation of Khovanov–Rozansky homology via categorical skew Howe duality**. *Advances in Mathematics*, 302 (2016), 1251–1339.
13. A. Lauda, H. Queffelec[†], and D.E.V. Rose. **Khovanov homology is a skew Howe 2-representation of categorified quantum $\mathfrak{sl}_m$** . *Algebraic and Geometric Topology*, 15-5 (2015), 2517–2608.
14. D.E.V. Rose. **A note on the Grothendieck group of an additive category**. *Bulletin of Chelyabinsk State University*, 2015. no. 3 (358). Mathematics. Mechanics. Informatics. Issue 17. pp. 135–139 (Proceedings of the *International Conference “Quantum topology”*).
15. D.E.V. Rose. **A categorification of quantum $\mathfrak{sl}_3$ projectors and the $\mathfrak{sl}_3$ Reshetikhin–Turaev invariant of tangles**. *Quantum Topology*, 5 (2014), no. 1, pp. 1–59.
16. D.E.V. Rose and I. Spitkovsky. **On the numerical range behavior under the generalized Aluthge transform**. *Linear and Multilinear Algebra*, vol. 56 no. 1&2 (January, 2008), pp. 163–177.
17. D.E.V. Rose and I. Spitkovsky. **On the stabilization of the Aluthge sequence**. *International Journal of Information and Systems Sciences - Special Issue on Matrix Analysis and Applications*, vol. 4 no. 1 (Spring, 2008), pp. 178–189.

Preprints:

18. E. Bodish, B. Elias, and D.E.V. Rose. **Type B webs**. 2026. *Preprint arXiv:2607.13252*. 65 pages.
19. E. Bodish, B. Elias, and D.E.V. Rose. **Spin Link Homology**. 2024. Submitted for publication. *Preprint arXiv:2407.00189*. 137 pages.
20. M. Hogancamp, D.E.V. Rose, and P. Wedrich. **Bordered invariants from Khovanov homology**. 2024. Submitted for publication. *Preprint arXiv:2404.06301*. 58 pages.
21. H. Queffelec, D.E.V. Rose, and A. Sartori. **Annular Evaluation and Link Homology**. 2018. *Preprint arXiv:1802.04131*. 47 pages.

In preparation:

22. M. Hogancamp, D.E.V. Rose, and P. Wedrich. **Heegaard Khovanov homology**. *In preparation* (arXiv ETA: 2026). Currently 27 pp.

GRANTS

To date, I have been awarded \$557,000 in grant funding:

- **AMS Young Scholars Grant** to fund *Girls Talk Math* in Summer 2026, PI: David Rose (percent effort: 0%), Award Amount \$11,000 (direct cost: \$11,000), Dates: 4/24/2026 – 8/15/2026
- **NSF CAREER Grant DMS-2144463**: *Link homology – in type A and beyond*, PI: David Rose (percent effort: 16.67%), Award Amount \$425,000 (direct cost: \$285,370), Dates: 7/1/2022 – 6/30/2027
- **Simons Collaboration Grant**: *Research on knot invariants, representation theory, and categorification*, PI: David Rose (percent effort: 0%), Award Amount: \$42,000, Direct Cost: \$35,000, Dates: 9/1/2017 – 8/31/2022
- **UNC Junior Faculty Development Award**, PI: David Rose (percent effort: 8.88%), Award Amount: \$10,000, Dates: 1/1/2019 – 12/31/2019
- **NSA Young Investigator Grant**: *Dualities in higher representation theory and low-dimensional topology*, PI: David Rose (percent effort at UNC: 8.33%), Award Amount: \$40,000, Direct Cost (at UNC): \$17,391, Dates: 1/21/2016 – 12/31/2017
- **AMS Simons Travel Grant**, PI: David Rose, Award Amount: \$4,000, Dates: 7/1/2015 – 12/31/2015
- **Zumberge Fund Individual Grant Award**: *Categorified quantum groups and knot homology*, PI: David Rose, Award Amount: \$25,000, Dates: 7/1/2013 – 6/30/2014

CONFERENCE AND SEMINAR TALKS

1. University of North Carolina Wilmington, Algebra and Number Theory Seminar, *TBD*, February 8, 2027 (forthcoming)
2. IPAM Workshop on Categorification in Quantum Topology, *TBD*, November 16–19, 2026 (forthcoming)
3. Fall AMS Western Sectional Meeting special session on Recent Developments in Tensor Categories and Their Applications, Arizona State University, *Type B Webs*, November 7–8, 2026 (forthcoming)
4. IPAM Tutorial on Foams, Khovanov homology, and skein lasagna modules, *Webs, foams, and invariants in low-dimensional topology*, September 24, 2026
5. University of North Carolina at Chapel Hill, Mathematics Colloquium, *On the categorification of 3d TQFTs*, August 27, 2026
6. Universität Hamburg, Quantum Topology and Categorification Seminar, *Folding foams*, May 20, 2026
7. ICERM Workshop on Webs in Algebra, Geometry, Topology and Combinatorics, *Type B webs and spin link homology*, December 12, 2025
8. Categorification in Low Dimensional Topology, Bochum, Germany, *Towards a categorification of the Turaev–Viro TQFT* (four 1.5-hour lectures), July 22–25, 2025

9. Columbia University, Geometric Topology Seminar, *Towards a higher TQFT from Khovanov homology*, March 7, 2025
10. University of North Carolina at Chapel Hill, Topology Seminar, *Spin link homology*, September 24, 2024
11. Stanford University, Topology Seminar, *Towards a categorification of the Turaev–Viro TQFT*, June 4, 2024
12. Uppsala University, Algebra Seminar, *Spin link homology*, May 21, 2024
13. UC Berkeley, String-Math Seminar, *Towards a categorification of the Turaev–Viro TQFT*, April 15, 2024
14. Michigan State University, Topology Seminar, *A Kirby color (or two) for Khovanov homology*, September 19, 2023
15. Michigan State University, RTG Seminar, *The Temperley–Lieb and Bar–Natan categories*, September 18, 2023
16. University Quantum Symmetries Lectures (UQSL), *A Kirby color for Khovanov homology*, January 12, 2023
17. QUAntum groups, Categorification, Knot invariants, and Soergel bimodules II, University of Oregon, *A Kirby color for Khovanov homology*, August 11, 2022
18. University of North Carolina at Chapel Hill, Geometric Methods in Representation Theory Seminar, *Type C Webs*, April 29, 2022
19. Categorical Methods in Representation Theory and Quantum Topology, University of Virginia, *Type C Webs*, April 17, 2022
20. Triangle Area Graduate Mathematics Conference (plenary faculty lecture), North Carolina State University, *Quantum knot invariants and webs*, April 2, 2022
21. North Carolina State University, Algebra and Combinatorics Seminar, *Type C Webs*, March 28, 2022
22. Penn State University, Colloquium, *Link Homology*, March 21, 2022
23. American Institute of Mathematics, Fundamentals of Link Homology Seminar, *Triply-graded link homology, Soergel bimodules, (and Traces)*, August 4, 2021
24. American Institute of Mathematics, Fundamentals of Link Homology Seminar, *Basics of Khovanov homology*, June 30, 2021
25. Spring AMS Western Sectional Meeting special session on Diagrammatic and Combinatorial Methods in Representation Theory, *Type C Webs*, May 1, 2021
26. UC Davis, Algebra & Discrete Mathematics Seminar, *Type C Webs*, March 11, 2021
27. Fall AMS Western Sectional Meeting special session on Monoidal Categories in Representation Theory, University of Utah, Salt Lake City, *Webs in Type C*, October 25, 2020
28. QUAntum groups, Categorification, Knot invariants, and Soergel bimodules, University of Oregon, *Webs in Type C*, August 12, 2020
29. Spring AMS Southeastern Sectional Meeting special session on Categorical Representation Theory and Beyond, University of Virginia, *On the quantum type C spider*, March 15, 2020 (Cancelled due to COVID-19 pandemic)
30. ICERM Workshop on Illustrating Number Theory and Algebra, Brown University, *Webs, foams, knot invariants, and representation theory*, October 21, 2019

31. Workshop on Quiver Hecke algebra and its applications to topology, Nagoya, Japan, *Link homology via the categorified skein module of the annulus*, July, 8, 2019
32. UC Berkeley, String-Math Seminar, $\mathfrak{gl}_n$ homologies, annular evaluation, and symmetric webs, October 8, 2018
33. Interactions of low-dimensional topology and “higher” representation theory, Universität Zürich, Switzerland, $\mathfrak{gl}_n$ homologies, annular evaluation, and symmetric webs, September 19, 2018
34. CQ3MI workshop, University of Southern California, $\mathfrak{gl}_n$ homologies, annular evaluation, and symmetric webs, July 19, 2018
35. Categorification and Higher Representation Theory, Institute-Mittag Leffler, Djursholm, Sweden, $\mathfrak{gl}_n$ homologies, annular evaluation, and symmetric webs, July 10, 2018
36. Quantum Knot Homology and Supersymmetric Gauge Theory, Aspen Center for Physics, *Annular evaluation and link homology*, March 5, 2018
37. Mathematical Conference of the Americas, Special Session on Symmetry in Algebra, Topology, and Physics, Montreal, Canada, *Annular evaluation and link homology*, July 26, 2017
38. Quantum topology and categorified representation theory, Isaac Newton Institute, Cambridge University, *Traces, current algebras, and link homologies*, June 26, 2017
39. Triangle Area Graduate Mathematics Conference (plenary faculty lecture), Duke University, *Knot invariants and dualities*, April 23, 2017
40. North Carolina State University, Geometry and Topology Seminar, *Quantum knot and link invariants from the symmetric perspective*, March 1st, 2017
41. Institut Montpellierain Alexander Grothendieck, Séminaire Topologies, *Quantum knot and link invariants from the symmetric perspective*, September 22, 2016
42. University of North Carolina at Chapel Hill, *Link homology via traces, current algebras, and dualities*, February 17, 2016
43. University of North Carolina at Chapel Hill, Mathematics Colloquium, *Low-dimensional topology, representation theory, and categorification*, February 16, 2016
44. Dartmouth College, Mathematics Colloquium, *Low-dimensional topology, representation theory, and categorification*, February 9, 2016
45. Joint Mathematics Meeting, AMS special session on Geometric and categorical methods in representation theory, Seattle, Washington, *Current algebras, Khovanov-Rozansky homology, and annular link invariants*, January 8, 2016
46. Joint Mathematics Meeting, AMS special session on Topological Representation Theory, Seattle, Washington, *Howe dualities and link invariants*, January 6, 2016
47. Knots in Washington – Plenary Talk, George Washington University, *Khovanov-Rozansky homology and current algebras*, December 4th, 2015
48. Duke University, Geometry/Topology Seminar, *Quantum knot invariants and Howe dualities*, April 7th, 2015
49. North Carolina State University, Geometry and Topology Seminar, *Quantum knot invariants and Howe dualities*, April 6th, 2015

50. University of North Carolina at Chapel Hill, Physically Inspired Mathematics Seminar, *Annular Khovanov-Rozansky homology*, April 2nd, 2015
51. University at Buffalo (SUNY), Mathematics Colloquium, *Categorification in topology and representation theory*, December 8th, 2014
52. The Joint Los Angeles Topology Seminar, UCLA, *Annular Khovanov homology via trace decategorification*, December 1, 2014
53. UC Davis, Algebra & Discrete Mathematics Seminar, *Khovanov-Rozansky homology via categorified quantum groups and skew Howe duality*, November 24, 2014
54. Fall AMS Southeastern Section Meeting special session on Algebraic structures motivated by Knot Theory, University of North Carolina at Greensboro, *Annular Khovanov homology via trace decategorification*, November 8, 2014
55. Centre for quantum geometry of moduli spaces (QGM), Aarhus, Denmark, *From the Jones polynomial to Khovanov-Rozansky homology via skew Howe duality*, July 30, 2014
56. NSF/CBMS Regional Conference on Higher Representation Theory, North Carolina State University, *From the Jones polynomial to Khovanov-Rozansky homology via skew Howe duality*, July 7, 2014
57. Centre de recherches mathématiques workshop on Categorification and geometric representation theory, Montreal, *The $\mathfrak{sl}_n$ foam 2-category via skew Howe duality*, June 9, 2014
58. Australian National University, Algebra and Topology Seminar, *Quantum link invariants and skew Howe duality*, March 18, 2014
59. Institut de Mathématiques de Jussieu, Paris, *Khovanov homology via categorified quantum groups*, December 17, 2013
60. University of North Carolina, Physically Inspired Mathematics Seminar, *Quantum link invariants and (higher) representation theory via skew Howe duality*, March 22, 2013
61. George Washington University, Topology Seminar, *Quantum link invariants and (higher) representation theory via skew Howe duality*, March 21, 2013
62. UC Riverside, Topology Seminar, *Quantum link invariants and (higher) representation theory via skew Howe duality*, March 5, 2013
63. Caltech, Geometry and Topology Seminar, *Khovanov homology, categorified quantum groups, and skew-Howe duality*, January 18, 2013
64. Claremont Colleges, Topology Seminar, *Quantum link invariants and (higher) representation theory via skew Howe duality*, October 30, 2012
65. UCLA, Topology Seminar, *Foams, Khovanov Homology, and Categorical Skew Howe Duality*, October 10, 2012
66. University of Southern California, Geometry and Topology Seminar, *A categorification of quantum $\mathfrak{sl}_3$ projectors and the $\mathfrak{sl}_3$ Reshetikhin-Turaev invariant of tangles*, April 27, 2012
67. University of Virginia, Geometry Seminar, *A categorification of quantum $\mathfrak{sl}_3$ projectors and the $\mathfrak{sl}_3$ Reshetikhin-Turaev invariant of tangles*, February 7, 2012
68. Baton Rouge Young Topologists Research Retreat, Louisiana State University, *A categorification of quantum $\mathfrak{sl}_3$ projectors and the $\mathfrak{sl}_3$ Reshetikhin-Turaev invariant of tangles*, January 9, 2012

69. Rice University, VIGRE Topology Seminar, *On Bar-Natan's "Khovanov's homology for tangles and cobordisms"*, March 9, 2010
70. The eighth workshop on numerical ranges and numerical radii (WONRA), Universität Bremen, *On the numerical range behavior under the generalized Aluthge transform*, July 15, 2006

ADVISING AND TEACHING ACTIVITIES

Postdoctoral researchers supervised:

- Calvin McPhail-Snyder, Fall 2021–Fall 2022

Graduate students supervised:

- Terence Carey (PhD expected: Spring 2028)
- Logan Gray, PhD, Summer 2026
Dissertation: "Computations in spin link homology"
Placement: Teaching Assistant Professor at UNC Chapel Hill
Mentored Awards: Linker award, 2025
- Luke Conners, PhD, Spring 2025
Dissertation: "Colored torus link homology"
Placement: Postdoc at Universität Zurich
Mentored Awards: International Congress of Chinese Mathematicians GTA Silver Award, 2024; Summer Research Fellowship, 2023
Mentored Publications:
 - L. Conners, **Fray functors and equivalence of colored HOMFLYPT homologies**. *Quantum Topology*. (2025), published online first DOI 10.4171/QT/249 79 pages.
 - L. Conners, **Row-Column Mirror Symmetry for Colored Torus Knot Homology**. *Selecta Mathematica, New Series*. 30 (2024), no. 97. 86 pages.
- Andrew Adair, PhD, Spring 2024
Dissertation: "Categorification of braid group representations"
Placement: Mathematician at the Department of Defense
Mentored Awards: Royster Society of Fellows, 2023–2024; Dissertation Completion Fellowship, 2023–2024; Summer Research Fellowship, 2022
- Logan Tatham, PhD, Summer 2020
Dissertation: "On the quantum type C spider"
Placement: Mathematician at the Department of Defense
Mentored Awards: G. de B. Robinson Award of the Canadian Mathematical Society, 2024

Undergraduate students supervised:

- Gillian Taylor, Honors thesis (defended: March 31, 2022)
Thesis: "3-manifold invariants in the category $\mathbf{Web}(\mathfrak{sp}_4)$ "
- Abby Watkins, Honors thesis (defended: April 16, 2021)
Thesis: "The Temperley–Lieb Category and its Trace"
Placement: Indiana University PhD program in mathematics

- Lily Gergle, Honors thesis (defended: April 8, 2021 with Highest Honors)
Thesis: “Categorification techniques for the Temperley–Lieb category and $\mathbf{Web}(\mathfrak{sp}_4)$ ”
Placement: University of Cambridge (Part III) and University of Illinois Urbana-Champaign PhD program in mathematics
Mentored Awards: UNC Brauer Prize, 2021
- Dylan O’Connor, Honors thesis (defended: May 17, 2020)
Thesis: “An Introduction to the Volume Conjecture”
Placement: CUNY graduate center PhD program in mathematics

Courses Taught at UNC:

- Math (347) in Stockholm: Linear Algebra for Applications, Summer 2026 (14 students)
- Math 347: Linear Algebra for Applications, Spring 2026 (107 students)
- Math 534: Elements of Modern Algebra, Spring 2026 (23 students)
- Math 681: Introductory Topology, Fall 2025 (10 students)
- Math 296: Directed Exploration in Mathematics (Monoidal Category Theory), Summer 2025 (1 student)
- Math 776: Algebraic Topology, Spring 2025 (11 students)
- Math 381: Discrete Mathematics, Spring 2025 (39 students)
- Math 891: Categorification in Algebra and Topology, Fall 2024 (8 students)
- Math 89: First Year Seminar in Knot Theory, Spring 2024 (24 students)
- Math 534: Elements of Modern Algebra, Spring 2024 (25 students)
- Math 681: Introductory Topology, Fall 2023 (10 students)
- Math 550: Topology, Fall 2022 (23 students)
- Math 692H: Honors Thesis in Mathematics, Spring 2022 (1 students)
- Math 381: Discrete Mathematics, Spring 2022 (39 students)
- Math 691H: Honors Research in Mathematics, Fall 2021 (1 student)
- Math 550: Topology, Fall 2021 (28 students)
- Math 231: Calculus of Functions of One Variable I, Fall 2021 (138 students)
- Math 692H: Honors Thesis in Mathematics, Spring 2021 (2 students)
- Math 578: Algebraic Structures, Spring 2021 (30 students)
- Math 691H: Honors Research in Mathematics, Fall 2020 (2 students)
- Math 676: Modules, Linear Algebra, and Groups, Fall 2020 (12 students)
- Math 381: Discrete Mathematics, Fall 2020 (42 students)
- Math 296: Directed Exploration in Mathematics (Research on Quantum Invariants), Fall 2020 (1 student)

- Math 920: Seminar and Directed Readings (Derived Categories), Spring 2020 (2 students)
- Math 692H: Honors Thesis in Mathematics, Spring 2020 (1 student)
- Math 681: Introductory Topology, Spring 2020 (11 students)
- Math 534: Elements of Modern Algebra, Spring 2020 (40 students)
- Math 296: Directed Exploration in Mathematics (Category Theory), Spring 2020 (2 students)
- Math 920: Seminar and Directed Readings (Homological Algebra), Fall 2019 (2 students)
- Math 676: Modules, Linear Algebra, and Groups, Fall 2019 (15 students)
- Math 691H: Honors Research in Mathematics, Fall 2019 (1 student)
- Math (381) in Stockholm: Discrete Mathematics, Summer 2019 (9 students)
- Math 776: Algebraic Topology, Spring 2019 (10 students)
- Math 381: Discrete Mathematics, Spring 2019 (42 students)
- Math 891: Categorification in algebra and topology, Spring 2018 (8 students)
- Math 231: Calculus of Functions of One Variable I, Fall 2017 (148 students)
- Math 534: Elements of Modern Algebra, Spring 2017 (40 students)
- Math 920: Seminar and Directed Readings (Low-dimensional Topology), Spring 2017 (1 student)
- Math 231: Calculus of Functions of One Variable I, Fall 2016 (136 students)

Courses Taught at USC:

- Math 245: Mathematics of Physics and Engineering I, Fall 2015
- Math 225: Linear algebra and linear differential equations, Fall 2015
- Math 225: Linear algebra and linear differential equations, Spring 2015
- Math 226: Calculus III, Fall 2014 (2 sections)
- Math 435: Vector Analysis and Introduction to Differential Geometry, Spring 2014
- Math 226: Calculus III, Spring 2014
- Math 440: Topology, Fall 2013
- Math 225: Linear algebra and linear differential equations, Spring 2013
- Math 226: Calculus III, Fall 2012 (2 sections)

Courses Taught at Duke University:

- Math 490: Algebraic methods in knot theory, Spring 2012
- Math 105L: Laboratory Calculus and Functions I, Spring 2010
- Math 122: Introductory Calculus II, Fall 2008

Curriculum and program design:

- Founder and Program Leader for *UNC Math in Stockholm*, a study abroad opportunity in which students travel to Sweden to learn mathematics in a focused and immersed setting.
- I designed the *First year seminar in knot theory* (Math 89) at UNC, an inquiry-based seminar on elementary knot theory.
- I designed the graduate course Math 891, *Categorification in algebra and topology* at UNC. This course served as an introduction to two aspects of the categorification program: link homology theories and higher representation theory, focusing on the $\mathfrak{sl}_2$ case.
- I designed the undergraduate course Math 490, *Algebraic methods in knot theory* at Duke University, through the Bass Fellowship program. This course served as an introduction to knot theory, the Jones polynomial, and Khovanov homology.

PROFESSIONAL SERVICE

Within UNC Chapel Hill:

- Organizer for Mathematics Department Exhibitions at UNC Science Expo, Spring 2017 – present
- Founder and program leader of UNC Math in Stockholm program: 2018 – 2020, 2025 – present
- UNC Mathematics Department Diversity Community and Outreach Committee, Summer 2020 – present
- Faculty coorganizer for Girls Talk Math, Spring 2025 – present
- UNC Mathematics Undergraduate Advising Committee, Fall 2021 – Spring 2023, Spring 2025 – present
- UNC Mathematics Tenure-Track Mentoring Committee (for Daping Weng), Fall 2024 – present
- Mathematics Comprehensive Exam Committees, 2019 – present
 - Geometry/Topology: Summer 2020 – Summer 2021, Summer 2022 – present
 - Algebra: Summer 2019 – Summer 2020, Summer 2021 – Summer 2022
- UNC Chancellor's Science Scholars program
 - Faculty interviewer: 2025, 2026
 - Research mentor: 2020–2021
- J. Burton Linker Award Selection Committee, 2024, 2025
- UNC organizer of the Triangle Topology Seminar, Fall 2016 – Fall 2019, Fall 2024 – Spring 2025
- Organizer for UNC Topology Seminar, Fall 2024
- UNC Mathematics Colloquium Committee, Summer 2021 – Spring 2023
- UNC Mathematics Tenure-track hiring committee, Fall 2022
- UNC Computation Medicine Tenure-track hiring committee, Spring 2022
- Course coordinator for UNC Math 231, Fall 2021
- UNC Mathematics Advising Policy Committee, Spring 2018 – Fall 2018

- Organizer for Topology Reading Group on Equivariant Cohomology, Fall 2017
- UNC Mathematics Postdoctoral Hiring Committee, Spring 2017
- UNC Mathematics Calculus Reform Working Group, Spring 2016 – Fall 2017
- Triangle Area Graduate Mathematics Conference Job Panel, December 3, 2016
- UNC Association for Women in Mathematics Academic Job Search Panel, October 26, 2016

Beyond UNC:

- Grant review for NSF Algebra and Number Theory program (forthcoming)
- Referee and/or quick opinion for:
 - *Advances in Mathematics*
 - *Algebraic and Geometric Topology*
 - *Annales scientifiques de l'École normale supérieure*
 - *Compositio Mathematica*
 - *Fundamenta Mathematicae*
 - *Geometry and Topology*
 - *International Mathematics Research Notices*
 - *Inventiones Mathematicae*
 - *Journal of the London Mathematical Society*
 - *Mathematische Annalen*
 - *Mathematische Zeitschrift*
 - *Quantum Topology*
 - *Selecta Mathematica*
- Coörganizer of the American Institute of Mathematics reading group *Fundamentals of Link Homology*, Summer and Fall 2021
- Grant reviewer for Simons Foundation, Spring 2019
- Organizer for USC Geometry/Topology Seminar, August 2012 – Spring 2015
- USC Geometry/Topology qualifying exam committee, Fall 2013 – Fall 2015
- Coörganizer of the *Workshop on Categorification and Representation Theory*, Los Angeles, CA, October 30–31, 2014
- Mentor for Augustus Hawkins High School (South Central Los Angeles) mathematical game day at USC, Spring 2014
- Coörganizer of the special session *Categorification in representation theory* at the AMS Fall sectional meeting, Riverside, CA, November 2–3 2013
- Organizer for USC graduate Topology reading group on Characteristic Classes, Spring 2013
- Organizer for Duke Mathematics Department Graduate/Faculty Seminar, January 2010 – December 2011
- Coörganizer and Lecturer for Duke Mathematics PhD written qualifying exam bootcamp, Summer 2010 and Summer 2011